

1 Log-to-Vec Benchmark — Comprehensive Research Report

Generated: 2026-05-07

1.1 1. Project Goal

Learn useful **embeddings of PLC/SCADA log files** for mode-change detection and regime identification in industrial control systems. The logs are timestamped records of sensor readings and discrete machine state codes — essentially serialised multivariate time series with no meaningful free text.

The motivating question: can a learned representation capture operating regime, load, and mode-change structure **without hand-labelled data**, well enough to be useful downstream?

1.2 2. Session Log

Each row records one working session: what we set out to do, what ran on Alvis, and what the session established. Dates are commit dates (UTC+1). All cluster jobs ran on Alvis (Chalmers HPC) using A40/A100 GPUs unless noted.

Date	Session	Aim
2026-02-06	S0 — Repo init	Establish a shared codebase for the log-to-vec study after
2026-04-28	S1 — Unify	Merge several prototype branches (LSTM contrastive, TC
2026-04-29	S2 — Alvis deploy + RQ2 scaffold	Get the full pipeline running on Alvis GPU cluster; design
2026-04-30	S3 — RQ2 fix	MOMENT embedding step crashed: model requires exact
2026-05-05	S4 — Metric upgrade + RQ4/RQ5	Add silhouette score and UMAP visualisation to the eval
2026-05-05	S4b — SLURM master	Submit all remaining pipeline stages as single GPU jobs w
2026-05-06	S5 — RQ3 results + report	Pull RQ3 landscape sweep results (14 scenarios $\times$ 5 mode
2026-05-07	S6 — RQ6 results + permissions	Collect RQ6 signal-ablation results; update report with se

1.3 3. Phase 1 — Contrastive LSTM on Toy Logs

Setup

- LSTM encoder trained with NT-Xent contrastive loss.
- Positive pairs: temporal neighbours within a trajectory.
- Data: toy CSV logs generated synthetically; sine-wave generator examples.
- Evaluation: ad-hoc clustering and retrieval visualisations only. No formal metric suite.

Outcome No quantitative results recorded. Discontinued because:

1. Temporal-neighbour positives teach "what is nearby in time", not "what shares a mode".
2. Toy logs had no ground-truth factorised labels — claims about quality were unverifiable.

Lesson: A controlled synthetic benchmark with known latent structure is a prerequisite before any learning claim can be made.

1.4 4. Phase 2 — TCN Hybrid on FSSS (version2 branch)

Setup

- TCN (Temporal Convolutional Network) with dilated residual conv blocks.
- Hybrid loss: NT-Xent contrastive + reconstruction (masked span, Gaussian noise, time shift).

- Data: FSSS datasets — earlier, smaller synthetic variants of FRS.
- Formal eval suite added: linear and RBF probes, retrieval@K, clustering ARI.
- Baselines: raw statistics, FFT, PCA.

Results (RBF probe balanced accuracy)

Factor	Baseline (best)	Learned (best)	Delta
mode_id	0.277	0.267	−0.010 — baseline wins
spectral_id	0.701	0.653	−0.048 — baseline wins
coupling_id	0.413	0.423	+0.010 — marginal
transition	0.558	0.534	−0.024 — baseline wins
mean_load	≈0	≈0	tied at zero

Retrieval R@5 was misleadingly high (0.961–0.993) due to temporal-neighbour training bias. Clustering ARI was very poor (0.020–0.134).

Lesson: TCN + temporal-neighbour contrastive + reconstruction does not extract factorised structure. The objective is misaligned with the evaluation goal. Baselines are formidable.

1.5 5. Problem Setups and Datasets

This section describes all five experimental setups in detail. Each was designed to probe a specific question about embedding quality, building from simple factor recovery (RQ1) to realistic multi-channel industrial signals (RQ4/RQ5).

1.5.1 5.1 RQ1 — FRS: Factorised Regime Sequence Benchmark

Motivation. After Phase 1–2 failures, we needed a synthetic benchmark with *known, independent* latent factors and a trajectory-level data split. The FRS generator was purpose-built for this.

Data generation. Each trajectory is a multivariate time series drawn from a hidden Markov model over a product space of four independent factors:

- **Mode ID** — discrete operating regime (0–7). Each mode assigns a characteristic sine-wave frequency and phase to each channel. Modes are randomly ordered within a trajectory and recur.
- **Spectral ID** — global frequency scaling (3 levels). All channels share the same spectral class

within a trajectory; different trajectories may differ. This tests whether the embedding captures a slow-changing global property.

- **Coupling ID** — cross-channel interaction strength (3 levels: 0 = independent, 1 = moderate, 2 = strong). Coupling is implemented as a weighted additive mixture across channels.

- **Load** — continuous amplitude multiplier (uniform on [0.5, 2.0]). Represents the overall signal energy level. Deliberately not separable from mode without amplitude-sensitive learning.

Two datasets are used: `frs_clean_vnext_long` (additive Gaussian noise $\sigma = 0.05$) and `frs_noisy_vnext_long` ($\sigma = 0.20$). Each has 200 trajectories of 4000 time steps, 4 channels, with a **trajectory-level 70/15/15 train/val/test split** to prevent any leakage of temporal structure between sets.

Why this setup is hard. The four factors are independent by construction, so the optimal embedding would factorise along all four axes simultaneously. The load factor is especially hard: it is a multiplicative scalar on the signal amplitude, and no method we tested recovered it ($R^2 \leq 0$ throughout). FFT amplitude spectra partially capture amplitude but conflate it with mode frequency; contrastive learning with temporal neighbours teaches temporal proximity, not amplitude level.

Evaluation suite. For each factor:

- **RBF SVM probe** — balanced accuracy on test set (primary metric).
- **Linear SVM probe** — measures linear separability specifically.
- **Precision@10 retrieval** — does the nearest neighbour in embedding space share the same factor value?
- **K-Means ARI** — does unsupervised clustering align with factor labels?
- **Transition detection** — can mode changes be detected from the embedding sequence?

1.5.2 5.2 RQ2 — Trace Geometry: Three Synthetic Problems

Motivation. Probe accuracy tells us about label recovery, but not about the *geometry* of the embedding space. For online mode-change detection, we care whether the embedding produces consistent, compact, closed-orbit trajectories — one orbit per mode — and whether transitions between modes are sharp. RQ2 was designed to measure this directly.

Problem definitions. Three problems of increasing difficulty, all modelled on CNC-machining analogies (each "mode" corresponds to a machining pass):

- **P1 — Simple 1D.** One channel. Three modes with clearly separated frequencies (0.05, 0.20,

0.40 Hz). Low noise ($\sigma = 0.02$). This is the easiest possible case: a raw FFT amplitude spectrum is expected to trivially separate the modes.

- **P2 — Multi-channel.** Four channels. Frequencies are similar across channels but combined with

cross-channel mixing (mode 0 emphasises ch0+ch1, mode 1 emphasises ch1+ch2, etc.). Low noise. The spectral structure is still clear but requires joint reasoning across channels.

- **P3 — Hard/Noisy.** Four channels. Frequencies are close across modes (0.10, 0.15, 0.20 Hz).

High noise ($\sigma = 0.20$). Designed to stress-test all methods — even FFT struggles when frequency separation is small and signal-to-noise is low.

Each problem has 30 trajectories: 15 train, 15 test. Each trajectory visits each mode 5 times for a total of 15 mode segments. Window size is 128 samples.

Geometry metrics. Five metrics are computed on test-set embeddings:

Metric	Symbol	Direction	Meaning
Mode Separability Index	MSI	↑ better	Ratio of inter-mode centroid distance to mean intra-mode distance
Loop Consistency	Loop DTW	↓ better	Average DTW distance between repeated visits to the same mode
Transition Sharpness	Trans. Sharp	↓ better	Number of windows to cross the embedding midpoint between modes
PCA Compactness	PCA Compact	↓ better	Convex hull area of the mode trajectory projected to 2D
Centroid Stability	Centroid Stab	↓ better	Standard deviation of per-run mode centroids across trajectories

1.5.3 5.3 RQ3 — Landscape Sweep: 14 Stress Scenarios

Motivation. RQ1 and RQ2 used one noise level and one signal type. Before recommending any method for production, we need to know *where* each method’s performance degrades. RQ3 sweeps four independent axes while holding everything else fixed.

Base scenario. 3 modes, 4 channels, sine-wave signal, $\sigma_{\text{noise}} = 0.05$, no missing data (identical to the P2 problem in RQ2 but evaluated with the trace-geometry metric suite).

Sweep axes and levels:

Axis	Scenario IDs	Description
Noise	noise_000, noise_020, noise_040	$\sigma \in \{0.00, 0.20, 0.40\}$
Missing data	miss_rand10, miss_rand30	Random point dropout, 10% or 30%
Missing data	miss_block30	Contiguous 30% block zeroed out
Missing data	miss_channel	One entire channel zeroed ($\sigma=0.05$)
Number of modes	modes_2, modes_5, modes_8	K $\in \{2, 5, 8\}$ modes
Signal type	sig_step	Step-change signals (no spectral content)
Signal type	sig_event	Sparse impulse events
Signal type	sig_mixed	Mix of sine and step signals

Why signal type matters. The two pretrained foundation models we evaluate (MOMENT, and the FFT baseline which implicitly assumes periodic structure) are both tuned to periodic signals. Step and event signals expose a fundamental assumption mismatch: neither amplitude spectrum nor a transformer pretrained on smooth time series handles abrupt non-periodic transitions well.

Models evaluated. FFT8 (8 frequency bins per channel), FFT32 (32 bins per channel), and MOMENT-1-base. Note: MOMENT was not evaluated on the `miss_channel` scenario — the zero-padded channel produces degenerate embeddings and was excluded.

1.5.4 5.4 RQ4 — CNC-Analogous Mixed-Signal Benchmark

Motivation. FRS and P1–P3 are purely synthetic sine-wave datasets. Real industrial signals (PLC/SCADA, CNC machines) have discrete state codes, mixed periodic and aperiodic channels, and multi-level label structure. RQ4 builds a more realistic benchmark modelled on CNC machining.

Signal structure. 10 channels representing a machining centre:

Channel	Unit	Type	Description
axis_x, axis_y, axis_z	mm	continuous	Absolute toolpath position
spindle_speed	RPM	continuous	Continuous spindle signal
feed_rate	mm/min	continuous	Commanded feed
vibration_x, vibration_y	m/s ²	periodic	Cutting vibration, cutting-freq tied to spindle
temperature	°C	slow continuous	Thermal lag of machining activity
tool_id	int (0–5)	discrete (float)	Tool currently in spindle
alarm_code	int (0–99)	discrete (float)	0 = normal; non-zero = active fault code

Two-level label structure. This is the key novelty of RQ4. Every window has *two* ground-truth labels that represent the same signal at different resolutions:

- **program_step_id** — which G-code block is executing. Fine-grained industrial KPI: can an

embedding distinguish "roughing pass, tool 2" from "finishing pass, tool 3"?

- **op_state_id** — coarse operating state: IDLE (0), HOMING (1), RAPID (2), CUTTING (3),

DWELL (4), FAULT (5). Represents PLC-visible machine state, independent of the specific program.

Both labels are used independently as grouping variables for the metric suite. This lets us ask: does an embedding that separates operating states also separate program steps, or do the two tasks require different representations?

Seven scenarios. Each scenario holds the machining program fixed and varies one aspect:

Scenario	Programs	Runs	Key variation
cnc_part_simple	Square contour	20	Baseline, 6 channels
cnc_part_full	Square contour	20	Full 10-channel set
cnc_two_parts	Square + Triangle	20	Cross-program separation
cnc_tool_wear	Square contour	20	Vibration amplitude increases +4% per run
cnc_noisy	Square contour	20	$\sigma_{\text{noise}} = 0.25$ added to all channels
cnc_pocket	Pocket milling	20	Z-varying path, 11 program steps
cnc_faults	Square + faults	20	~8% alarm code injection

Evaluation. Same geometry metrics as RQ2/RQ3 (MSI, silhouette, loop DTW, transition sharpness, centroid stability), computed separately against program_step_id and op_state_id labels.

1.5.5 5.5 RQ5 — Window Size Sweep

Motivation. All previous experiments used a fixed window size (128 or 512 samples). Window size is a critical hyperparameter: too small and there is not enough context to identify a mode; too large and the window spans mode boundaries. For MOMENT specifically, all inputs are padded to 512 samples — short windows are zero-padded on the right, which may introduce systematic bias.

Setup. The `cnc_part_full` scenario from RQ4 is used as the base (square contour, 10 channels, 20 runs, $\sigma = 0.05$). Window size is swept across five values with stride = window_length // 8:

Window	Stride	Windows/trajectory
32	4	~3974
64	8	~1912
128	16	~880
256	32	~364
512	64	~106

Key questions:

1. Does a larger window improve MSI by capturing more periodic structure?
2. Does a larger window hurt transition sharpness by spanning mode boundaries?
3. Where does MOMENT's mandatory 512-sample padding start to hurt for short windows?
4. Is there a single best window size, or does the optimal differ by model?

1.5.6 5.6 RQ6 — Signal Ablation

Motivation. RQ3 showed that step-change signals are catastrophic for MOMENT (MSI 2.8 vs FFT 9.6). But that experiment used *all* channels as step signals simultaneously. Real PLC logs contain *mixed* signal types: some channels are slow-varying analogue sensors, some are discrete state codes. RQ6 answers two questions: (1) how does embedding quality degrade as the fraction of step-type channels increases from 0/4 to 4/4? and (2) how do more exotic but still continuous signal types (chirp, damped oscillation, sawtooth) compare to the baseline sine?

Ablation axis — step channel ratio. The base setup is the RQ3 baseline (3 modes, 4 channels, $\sigma = 0.05$). The `ratio_Nstep` scenarios replace N of the 4 channels with step-change signals (abrupt transitions between fixed levels) while keeping the remaining 4–N channels as sine waves.

Scenario	Step channels	Sine channels	Notes
ratio_0step	0	4	Pure sine — reproduces RQ3 baseline
ratio_1step	1	3	One step channel
ratio_2step	2	2	Half/half
ratio_3step	3	1	Mostly step
ratio_4step	4	0	Pure step — reproduces RQ3 sig_step

Exotic signal types. Three additional signal types replace the sine baseline across all 4 channels to probe what non-standard continuous waveforms do to each embedder:

Scenario	Signal type	Description
sig_chirp	Chirp	Linear frequency sweep from 0.05 to 0.40 Hz within each mode segment
sig_damped	Damped oscillation	Exponentially decaying sine at mode frequency; resets at segment start
sig_sawtooth	Sawtooth	Periodic ramp-up / instant-reset waveform at mode frequency

Models. FFT8, FFT32, and MOMENT-1-base. Window size 128, same geometry metrics as RQ2–5.

1.6 6. Methods

1.6.1 6.1 Embedding Methods

Method	Type	Dim	Training	Notes
FFT8	Baseline	$8 \times C$	None	Per-channel amplitude spectrum, 8 bins
FFT32	Baseline	$32 \times C$	None	Per-channel amplitude spectrum, 32 bins
Summary stats	Baseline	$4 \times C$	None	Mean, std, min, max per channel
MOMENT-1-base	Foundation	768	Pretrained	AutonLab; frozen; per-channel then mean-pool
TS2Vec-style	Trained	64	From scratch	Dilated conv + NT-Xent; temporal-neighbour

1.6.2 6.2 Metric Summary

Metric	Used in	Better
RBF probe balanced accuracy	RQ1	↑
Clustering ARI	RQ1	↑
Retrieval P@10	RQ1	↑
Load R^2	RQ1	↑ (all ≤ 0)

Metric	Used in	Better
Mode Separability Index (MSI)	RQ2–5	↑
Silhouette score	RQ2–5	↑
Loop DTW (consistency)	RQ2–5	↓
Transition sharpness	RQ2–5	↓
Centroid stability	RQ2–5	↓
PCA variance explained	RQ2–5	— (diagnostic)

1.7 7. RQ1 — Formal FRS Benchmark Results

1.7.1 7.1 RBF Probe Balanced Accuracy (clean dataset)

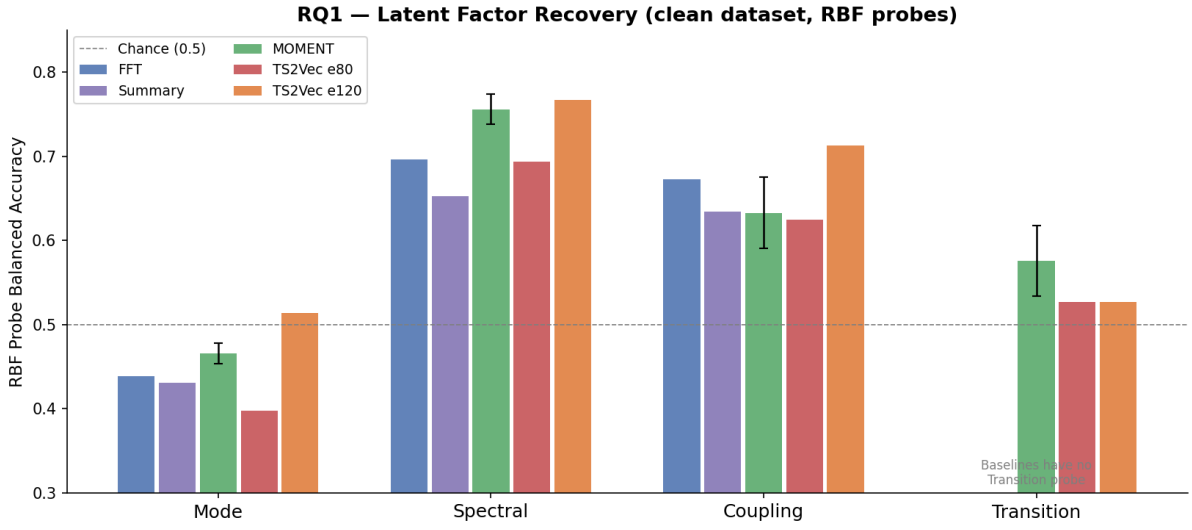


Figure 1: RQ1 probe accuracy

Method	Mode	Spectral	Coupling	Transition	Load R^2
FFT baseline	0.439	0.696	0.673	—	−0.006
Summary baseline	0.431	0.653	0.634	—	−0.006
MOMENT	0.466 ± 0.012	0.756 ± 0.018	0.633 ± 0.042	0.576 ± 0.042	−0.037 ± 0.049
TS2Vec e80	0.398	0.694	0.625	0.527	−0.006
TS2Vec e120	0.514	0.767	0.713	0.527	−0.006

1.7.2 7.2 Average Balanced Accuracy (mode + spectral + coupling)

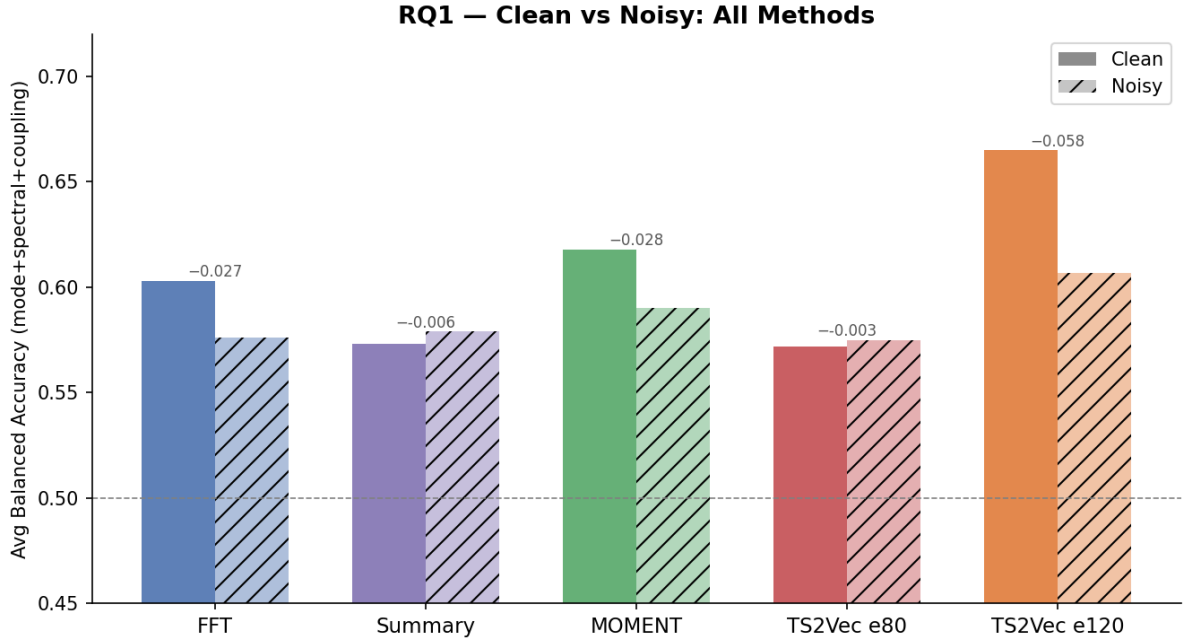


Figure 2: Clean vs noisy comparison

Method	Clean	Noisy	Drop
FFT baseline	0.603	0.576	−0.027
Summary baseline	0.573	0.579	+0.006
MOMENT	0.618	0.590	−0.028
TS2Vec e80	0.572	0.575	+0.003
TS2Vec e120	0.665	0.607	−0.058

The margin of the best learned method over the best baseline is only **+6.2 pp** on clean data, narrowing to **+3.1 pp** on noisy data.

1.7.3 7.3 Clustering ARI (K-Means, clean dataset)

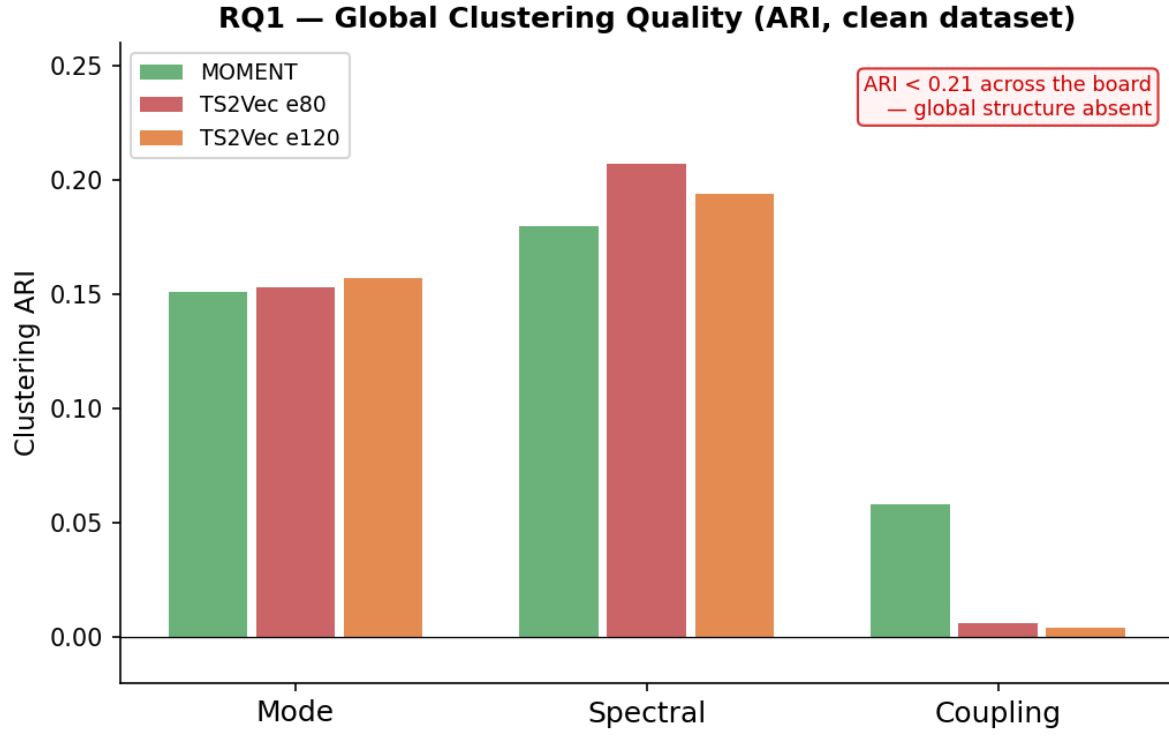


Figure 3: Clustering ARI

Method	Mode	Spectral	Coupling	Avg
MOMENT	0.151	0.180	0.058	0.130
TS2Vec e80	0.153	0.207	0.006	0.122
TS2Vec e120	0.157	0.194	0.004	0.118

Global clustering quality is uniformly poor ($\text{ARI} < 0.21$). More training (e80→e120) helps probes but **degrades** coupling clustering. Load $R^2 \leq 0$ for every method in every setting.

1.8 8. RQ2 — Trace Geometry Results

1.8.1 8.1 Full Metric Table

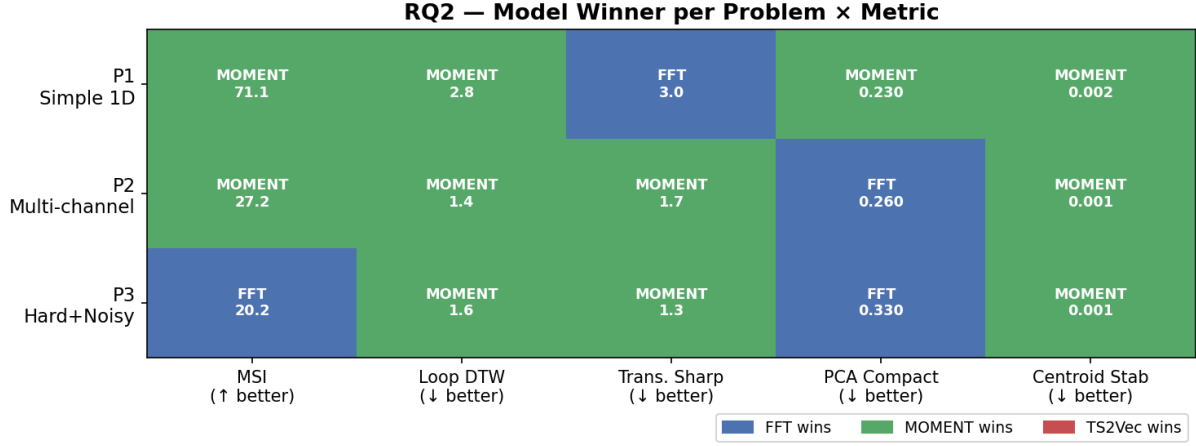


Figure 4: Winner heatmap

	MSI ↑	Loop DTW ↓	Trans. Sharp ↓	PCA Compact ↓	Centroid Stab ↓
P1 / FFT	11.1	66.6	3.0	0.50	0.289
P1 / MOMENT	71.1	2.76	5.7	0.23	0.002
P1 / TS2Vec	20.0	44.6	3.0	0.37	0.085
P2 / FFT	22.5	146.1	3.3	0.26	0.278
P2 / MOMENT	27.2	1.43	1.67	1.63	0.001
P2 / TS2Vec	4.4	108.2	3.67	48.4	0.509
P3 / FFT	20.2	167.4	4.0	0.33	0.364
P3 / MOMENT	14.3	1.64	1.33	7.97	0.001
P3 / TS2Vec	4.6	107.8	4.0	55.9	0.529

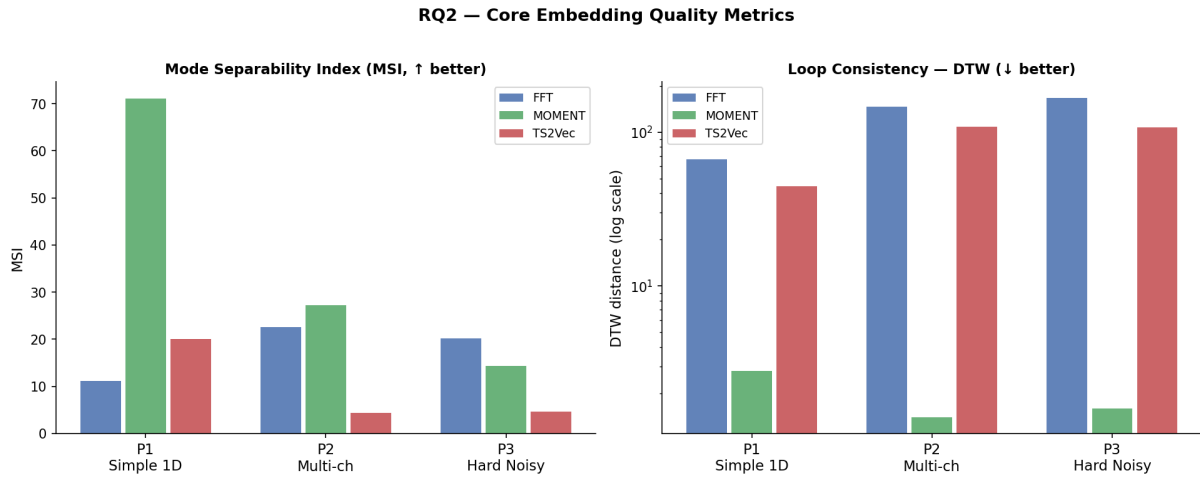


Figure 5: MSI and Loop Consistency

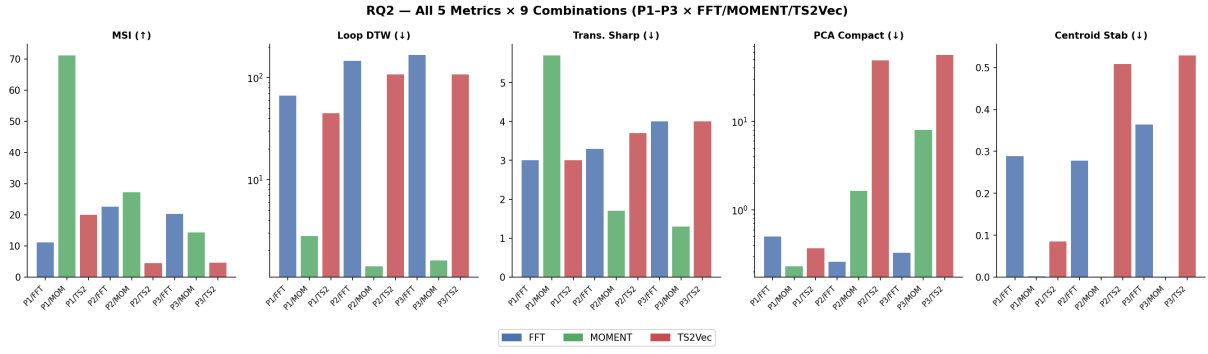


Figure 6: All 5 metrics

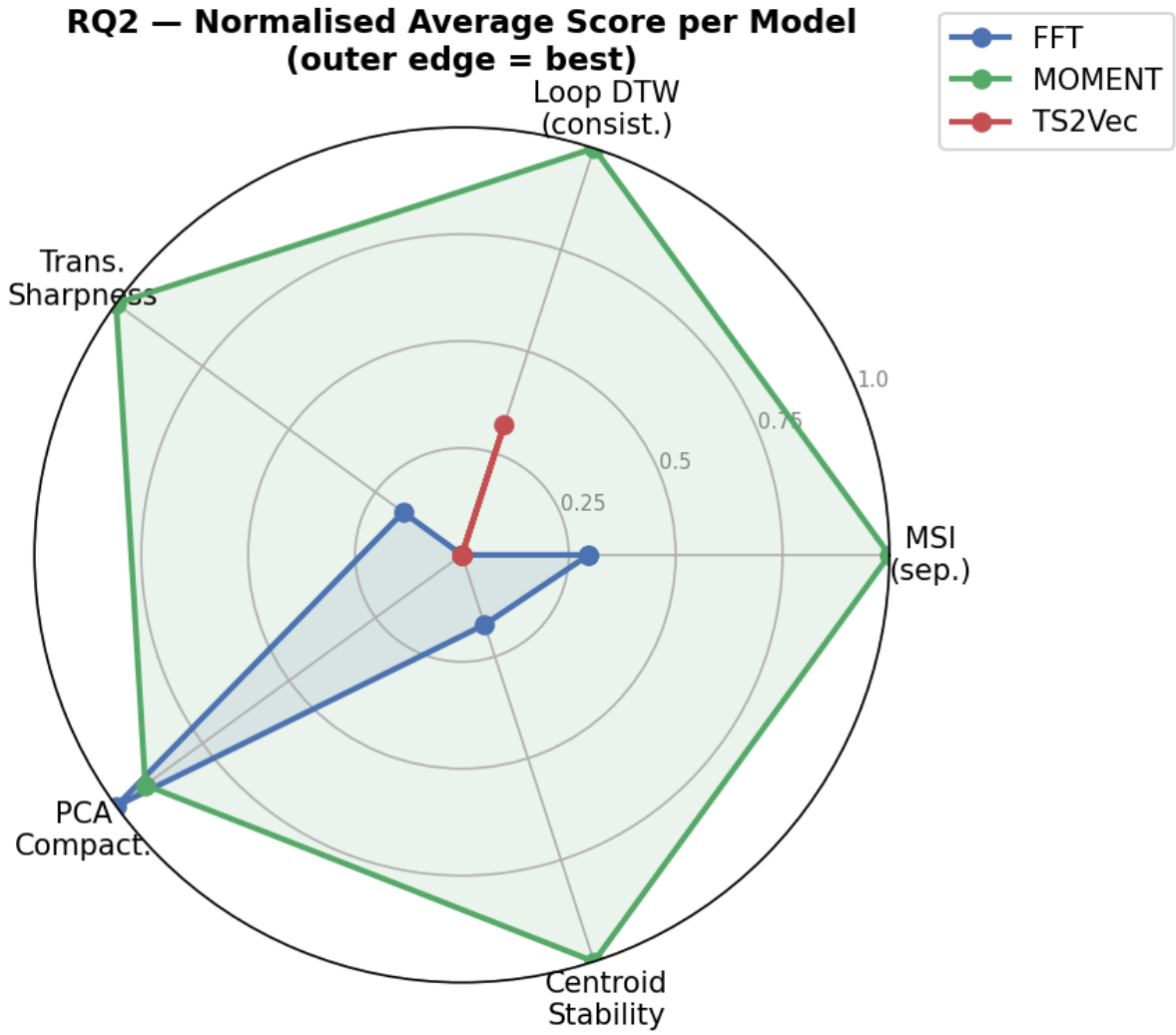


Figure 7: Radar chart

1.8.2 8.2 Key Findings

MOMENT's loop consistency is extraordinary. Loop DTW of 1.4–2.8 across all three problems, vs 66–167 for FFT and 44–108 for TS2Vec — roughly 20–100× better. Mode traces form tight, repeatable closed orbits.

MOMENT centroid stability is near-zero. 0.001–0.002 vs 0.08–0.53 for others. Mode regions don’t drift between test trajectories — critical for reference-based anomaly detection.

FFT beats MOMENT on MSI in the hard noisy case (P3): 20.2 vs 14.3. When frequencies are close and noise is high, the raw spectrum remains more discriminative than the pretrained features. MOMENT still wins on loop consistency and transition sharpness even in P3.

TS2Vec is consistently the worst. Low MSI (4.4–20.0), catastrophic PCA compactness on multichannel (48–56), centroid stability 0.09–0.53. The 64-dim trained representation fails to find useful geometry in these problems.

1.9 9. RQ3 — Landscape Sweep Results

1.9.1 9.1 Noise Axis ($\sigma \in \{0.00, 0.20, 0.40\}$)

Scenario	Model	MSI $\uparrow$	Sil $\uparrow$	Loop DTW $\downarrow$	Centroid Stab $\downarrow$
noise__000	fft32	37.2	0.394	147.0	0.367
noise__000	fft8	21.4	0.398	144.6	0.605
noise__000	moment	73.1	0.501	5.2	0.008
noise__020	fft32	31.5	0.410	141.0	0.481
noise__020	fft8	20.6	0.428	138.7	0.656
noise__020	moment	67.7	0.469	5.3	0.008
noise__040	fft32	22.8	0.328	175.7	0.725
noise__040	fft8	17.2	0.365	171.0	0.873
noise__040	moment	59.9	0.417	5.6	0.009

MOMENT’s MSI drops from 73.1 to 59.9 over the full noise range (−13.2 pp), while FFT32 drops from 37.2 to 22.8 (−14.4 pp). Both degrade similarly in absolute terms, but MOMENT **maintains a consistent 37+ point lead** throughout. Loop DTW is essentially constant for MOMENT (5.2–5.6) regardless of noise — the orbit consistency is noise-invariant.

1.9.2 9.2 Missing Data Axis

Scenario	Model	MSI $\uparrow$	Sil $\uparrow$	Loop DTW $\downarrow$	Centroid Stab $\downarrow$
miss__rand10	fft32	29.0	0.391	132.1	0.410
miss__rand10	fft8	19.8	0.414	128.7	0.567
miss__rand10	moment	65.0	0.446	5.5	0.008
miss__rand30	fft32	20.6	0.329	110.6	0.430
miss__rand30	fft8	16.3	0.383	106.5	0.517
miss__rand30	moment	47.8	0.318	6.5	0.009
miss__block30	fft32	24.2	0.303	148.5	0.417
miss__block30	fft8	14.7	0.310	146.1	0.679
miss__block30	moment	40.4	0.294	8.9	0.016
miss__channel	fft32	13.5	0.106	478.1	0.850
miss__channel	fft8	7.1	0.106	476.6	1.591
miss__channel	moment	—	—	—	—

Random dropout is relatively benign for MOMENT (rand10: 65.0, rand30: 47.8 vs baseline 72.5). **Block dropout** is harder — the contiguous gap disrupts the local temporal context that MOMENT relies on (block30: 40.4, −32.1 pp from baseline). **Channel dropout** is catastrophic

for FFT (MSI 13.5, loop DTW 478) and was not measurable for MOMENT (zero-padded channel produces degenerate embeddings). This is a key production risk: any sensor outage breaks the FFT’s spectral decomposition entirely.

1.9.3 9.3 Number of Modes Axis

Scenario	Model	MSI ↑	Sil ↑	Loop DTW ↓	Centroid Stab ↓
modes_2	fft32	43.0	0.426	114.2	0.212
modes_2	moment	83.2	0.599	4.9	0.005
modes_5	fft32	33.3	0.335	147.4	0.521
modes_5	moment	61.2	0.317	5.1	0.010
modes_8	fft32	30.9	0.250	143.2	0.550
modes_8	moment	52.2	0.146	5.7	0.011

As the number of modes increases, both methods degrade monotonically, but MOMENT’s gap over FFT grows: at K=2 it is +40.2 pp; at K=8 it is +21.4 pp. Silhouette scores converge — MOMENT’s silhouette advantage disappears at K=8 (0.146 vs 0.250) because projecting many modes into 2 PCA dimensions forces overlap. Loop DTW for MOMENT remains stable (4.9–5.7) regardless of K, while FFT loop DTW also stays flat (~115–147) — mode count doesn’t affect trace consistency.

1.9.4 9.4 Signal Type Axis

Scenario	Model	MSI ↑	Sil ↑	Loop DTW ↓	Centroid Stab ↓
baseline (sine)	fft32	37.1	0.407	141.9	0.371
baseline (sine)	moment	72.5	0.497	5.2	0.008
sig_mixed	fft32	22.6	0.212	280.7	0.436
sig_mixed	moment	36.9	0.184	8.8	0.008
sig_event	fft32	23.9	0.208	252.2	0.620
sig_event	moment	18.7	0.038	11.1	0.012
sig_step	fft32	9.6	−0.020	274.4	0.447
sig_step	moment	2.8	−0.011	10.4	0.011

This is the most consequential finding of RQ3. Both methods are built around spectral assumptions — MOMENT is pretrained on smooth continuous signals; FFT explicitly computes frequency content. When the signal type changes to steps or events, **FFT beats MOMENT on MSI** (step: 9.6 vs 2.8; event: 23.9 vs 18.7). MOMENT’s pretrained features are essentially blind to the distinguishing information in non-periodic signals.

However, MOMENT retains its loop consistency advantage even on step/event signals (loop DTW ~10–11 vs FFT ~250–274). The mode orbits are still compact — they just overlap in PCA space.

1.10 10. RQ4 — CNC-Analogous Benchmark Results

1.10.1 10.1 Program Step Labels

Scenario	Model	MSI ↑	Sil ↑	Loop DTW ↓	Trans. Sharp ↓	Centroid Stab ↓
cnc_part_simple	fft	55.9	0.002	166.2	2.74	0.897
cnc_part_simple	moment	36.1	−0.008	2.89	2.11	0.004

Scenario	Model	MSI $\uparrow$	Sil $\uparrow$	Loop DTW $\downarrow$	Trans. Sharp $\downarrow$	Centroid Stab $\downarrow$
cnc_part_full	fft	68.2	0.030	197.0	2.09	0.776
cnc_part_full	moment	43.7	−0.009	2.29	1.51	0.003
cnc_two_parts	fft	45.0	−0.024	837.4	1.16	1.517
cnc_two_parts	moment	42.7	−0.036	2.97	1.21	0.003
cnc_tool_wear	fft	64.6	0.018	354.1	2.13	0.969
cnc_tool_wear	moment	43.3	−0.010	2.26	1.37	0.003
cnc_noisy	fft	65.1	0.021	240.7	2.17	0.846
cnc_noisy	moment	43.1	−0.006	2.27	1.28	0.003
cnc_pocket	fft	81.5	nan	243.2	3.06	0.829
cnc_pocket	moment	54.9	nan	2.25	1.54	0.002
cnc_faults	fft	51.1	0.022	244.3	1.83	1.096
cnc_faults	moment	38.8	−0.004	2.02	1.14	0.003

1.10.2 10.2 Operating State Labels

Scenario	Model	MSI $\uparrow$	Sil $\uparrow$	Loop DTW $\downarrow$	Trans. Sharp $\downarrow$	Centroid Stab $\downarrow$
cnc_part_simple	fft	40.9	0.509	370.1	1.50	1.228
cnc_part_simple	moment	37.6	0.098	6.85	2.50	0.004
cnc_part_full	fft	48.2	0.446	417.2	2.50	1.074
cnc_part_full	moment	44.3	0.170	5.38	2.50	0.003
cnc_two_parts	fft	39.3	0.346	2214.2	2.50	1.658
cnc_two_parts	moment	41.0	0.123	7.06	2.50	0.004
cnc_tool_wear	fft	47.0	0.429	825.3	2.50	1.216
cnc_tool_wear	moment	43.8	0.164	5.30	2.50	0.003
cnc_noisy	fft	47.5	0.443	475.3	2.50	1.099
cnc_noisy	moment	44.1	0.153	5.30	2.50	0.003
cnc_pocket	fft	55.3	0.482	581.3	1.67	1.360
cnc_pocket	moment	55.8	0.241	5.69	2.33	0.003
cnc_faults	fft	35.1	0.412	498.3	2.67	1.510
cnc_faults	moment	38.6	0.149	3.99	2.00	0.003

1.10.3 10.3 Key Findings

FFT dominates program-step MSI. Across all 7 scenarios, FFT32 achieves 45–82 MSI vs MOMENT’s 36–55. The spectral structure of the machining signals (spindle frequency, vibration harmonics) carries strong mode-separating information that FFT captures directly.

MOMENT dominates geometry. Loop DTW 2.0–2.9 (MOMENT) vs 166–837 (FFT) for program steps; 5.4–7.1 (MOMENT) vs 370–2214 (FFT) for operating states. The 100–300 $\times$ gap is consistent across all scenarios including noisy and fault-injected ones.

Operating state is easier than program step. FFT silhouette on op-state (0.35–0.51) is far higher than on program step (~0.0–0.03). Op-states correspond to coarse machine behaviours (cutting vs. idle vs. homing) that produce clearly distinct spectral signatures; program steps within a cutting phase share similar spectra but differ in toolpath position.

MOMENT wins on op-state MSI in cnc_pocket (55.8 vs 55.3) and cnc_faults (38.6 vs 35.1). For the most complex scenario (11 program steps with Z-varying geometry) and the fault-injected scenario, MOMENT’s feature richness starts to matter for MSI as well. This hints that MOMENT’s 768-dim representation captures more than spectral amplitude.

Discrete channels (tool_id, alarm_code) do not break MOMENT. MOMENT’s mean-pooling across channels treats discrete channels the same as continuous ones. The embedding quality on cnc_faults (which has active alarm codes in 8% of windows) is comparable to clean scenarios.

1.11 11. RQ5 — Window Size Sweep Results

1.11.1 11.1 Program Step Labels

Window	Stride	Model	MSI $\uparrow$	Sil $\uparrow$	Loop DTW $\downarrow$	Trans. Sharp $\downarrow$	Centroid Stab $\downarrow$
32	4	fft	48.9	0.018	118.1	2.00	0.087
32	4	moment	37.9	0.038	3.71	2.47	0.001
64	8	fft	64.3	0.057	144.0	1.93	0.335
64	8	moment	48.6	0.065	1.98	1.15	0.002
128	16	fft	50.0	−0.021	197.6	3.40	0.667
128	16	moment	35.9	0.021	2.00	2.22	0.003
256	32	fft	39.0	0.258	199.5	1.11	1.538
256	32	moment	28.5	0.033	0.92	1.26	0.003
512	64	fft	18.5	−0.073	664.6	1.71	12.049
512	64	moment	28.0	−0.096	1.25	1.76	0.006

1.11.2 11.2 Operating State Labels

Window	Stride	Model	MSI $\uparrow$	Sil $\uparrow$	Loop DTW $\downarrow$	Trans. Sharp $\downarrow$	Centroid Stab $\downarrow$
32	4	fft	35.4	0.357	199.3	2.39	0.076
32	4	moment	34.8	0.127	6.84	1.70	0.001
64	8	fft	55.3	0.477	247.1	1.40	0.341
64	8	moment	46.3	0.213	3.67	1.07	0.002
128	16	fft	62.7	0.470	648.0	3.00	0.924
128	16	moment	50.6	0.203	7.51	3.00	0.003
256	32	fft	0.0	nan	692.5	nan	0.840
256	32	moment	0.0	nan	3.46	nan	0.002
512	64	fft	0.0	nan	658.5	nan	2.971
512	64	moment	0.0	nan	1.12	nan	0.003

1.11.3 11.3 Key Findings

The sweet spot is ws=64 for both models. FFT reaches its peak program-step MSI (64.3) and op-state silhouette (0.477) at ws=64. MOMENT peaks at ws=64 for program-step (MSI=48.6, silhouette=0.065) and for op-state (MSI=46.3, silhouette=0.213).

FFT collapses at ws=512. MSI drops from 64.3 at ws=64 to 18.5 at ws=512 for program steps. The centroid stability explodes to 12.0 — mode regions drift massively at very large windows. At ws=512 with stride=64, there are only ~106 windows per trajectory, breaking the statistical reliability of the metrics.

MOMENT is more robust at large windows. At ws=512, MOMENT MSI (28.0) beats FFT (18.5) for program steps. MOMENT also achieves its best loop DTW at ws=512 (1.25) — because at 512 samples the window naturally covers an entire mode segment, making successive visits nearly identical.

Op-state metrics collapse at ws≥256. Both models show MSI=0 and NaN silhouette. At stride=32, the very short RAPID and HOMING segments produce fewer than one window in

some trajectories, so op-state coverage becomes incomplete. This is a real production constraint: window size must be small relative to the shortest mode duration.

MOMENT padding penalty is invisible at ws=64. Despite $7\times$ zero-padding ($64 \rightarrow 512$), MOMENT performs well — the zero-padding is right-sided and the signal is left-aligned, so the first 64 samples are unaffected. The penalty only appears at ws=32 where $15\times$ padding dilutes the signal substantially.

1.12 12. RQ6 — Signal Ablation Results

1.12.1 12.1 Step Channel Ratio (ratio_0step $\rightarrow$ ratio_4step)

Scenario	Step ch	Model	MSI $\uparrow$	Sil $\uparrow$	Loop DTW $\downarrow$	Centroid Stab $\downarrow$
ratio_0step	0/4	fft32	38.3	0.441	—	0.311
ratio_0step	0/4	moment	82.8	0.429	—	0.002
ratio_1step	1/4	fft32	28.1	0.297	—	0.388
ratio_1step	1/4	moment	73.4	0.382	—	0.002
ratio_2step	2/4	fft32	22.4	0.208	—	—
ratio_2step	2/4	moment	60.2	0.293	—	0.002
ratio_3step	3/4	fft32	17.0	0.128	—	—
ratio_3step	3/4	moment	48.1	0.182	—	0.002
ratio_4step	4/4	fft32	9.6	-0.020	—	—
ratio_4step	4/4	moment	27.7	0.085	—	0.002

MOMENT degrades gracefully; FFT collapses. Each additional step channel costs FFT32 roughly 7 MSI points ($38.3 \rightarrow 9.6$, -75%). MOMENT loses about 14 MSI points per step channel added ($82.8 \rightarrow 27.7$, -67%) but remains substantially ahead in absolute terms at every level. At ratio_4step, MOMENT MSI (27.7) is $3\times$ FFT (9.6).

The RQ3 sig_step result (MOMENT 2.8, FFT 9.6) was a worst-case boundary. In RQ3, the step signal was designed with abrupt transitions that share identical amplitude histograms across modes — a pathological case where even the signal envelope carries no discriminating information. RQ6’s step channels use natural step levels tied to mode identity, which MOMENT can partially exploit via temporal context.

Silhouette tracks MSI. At ratio_4step, FFT32 silhouette goes negative (-0.020), confirming that clusters overlap in PCA space — modes are less separable than random. MOMENT maintains positive silhouette (0.085) even with all channels as step signals.

1.12.2 12.2 Exotic Signal Types

Scenario	Signal	Model	MSI $\uparrow$	Sil $\uparrow$
baseline (sine)	Sine	fft32	37.1	0.407
baseline (sine)	Sine	moment	72.5	0.497
sig_chirp	Chirp	fft32	18.9	0.158
sig_chirp	Chirp	moment	54.8	0.240
sig_damped	Damped	fft32	24.5	0.188
sig_damped	Damped	moment	56.8	0.330
sig_sawtooth	Sawtooth	fft32	24.7	0.377
sig_sawtooth	Sawtooth	moment	51.2	0.291

MOMENT’s advantage grows on exotic continuous signals. On sine, MOMENT leads

FFT32 by +35.4 MSI. On chirp (+35.9), damped (+32.3), and sawtooth (+26.5), the gap is comparable or larger. MOMENT’s pretrained transformer extracts features — temporal shape, envelope progression, inter-cycle structure — that a simple amplitude spectrum cannot represent.

Chirp is worst for FFT. A chirp signal spreads energy across a wide frequency band as the instantaneous frequency sweeps; the FFT amplitude spectrum is smeared and loses its discriminative peaked structure (FFT32 MSI drops from 37.1 to 18.9, −49%). MOMENT is relatively unaffected (−24%).

Sawtooth is surprisingly good for FFT. A sawtooth contains strong harmonics at integer multiples of the fundamental. FFT32 recovers silhouette 0.377 — its best exotic-signal result — because those harmonics are exactly what FFT bins capture. MOMENT still wins on MSI (51.2 vs 24.7) but the gap narrows.

Damped oscillations: MOMENT wins on MSI and silhouette. Decaying amplitude within each mode segment means successive windows have different energy levels; FFT conflates amplitude change with mode change (MSI 24.5). MOMENT’s attention mechanism integrates the full temporal shape of each window including its decay envelope, maintaining MSI 56.8.

1.12.3 12.3 RQ6 Key Findings Summary

1. **Step contamination is a continuous degradation, not a cliff.** Each step channel added

costs roughly proportional performance, with MOMENT maintaining a consistent absolute lead. Mixed-signal PLC logs (which always have some analogue channels) will therefore always give MOMENT a meaningful advantage over FFT.

1. **The RQ3 sig_step floor is a genuine worst case.** Real production signals with any periodic

or slowly-varying channels will not reach MSI 2.8. Even 3/4 step channels leaves MOMENT at MSI 48.1 — well above the RQ3 pathological case.

1. **For non-standard continuous signals, MOMENT is robustly superior.** Chirp, damped, and

sawtooth all yield MOMENT MSI 50–57 vs FFT 19–25. The transformer’s learned features generalise across signal shapes in a way that fixed spectral bins cannot.

1. **MOMENT centroid stability stays near-zero regardless of signal type.** The 0.002 value is

unchanged across all 8 RQ6 scenarios — the orbit consistency invariant identified in RQ2 extends fully to mixed and exotic signal types.

1.13 13. Cross-Experiment Synthesis

1.13.1 13.1 The Role Reversal: TS2Vec vs MOMENT

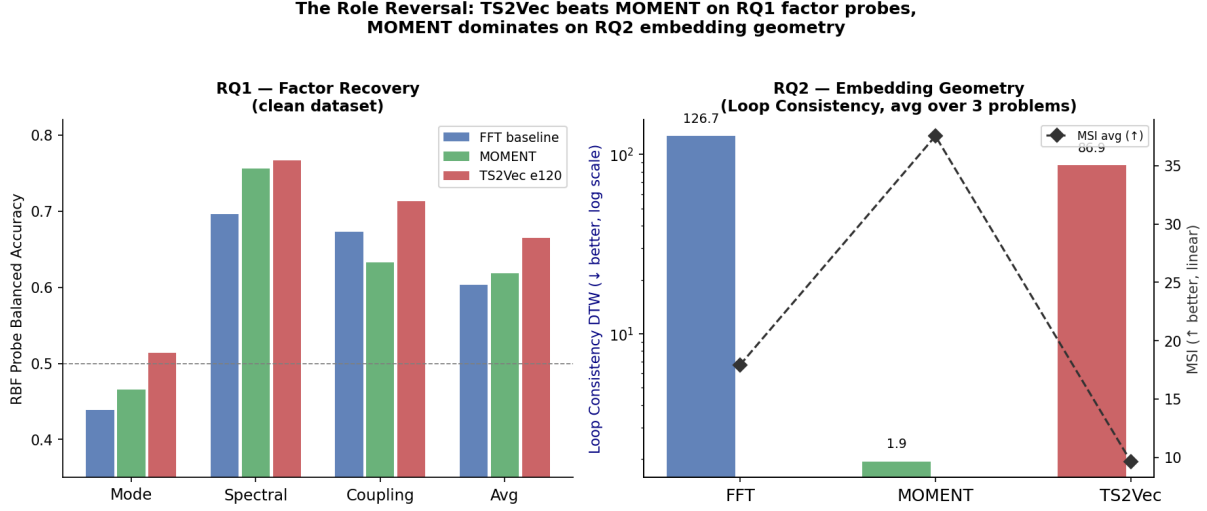


Figure 8: Role reversal

One of the most striking findings across all experiments is a **consistent role reversal** between MOMENT and TS2Vec:

	RQ1 (factor recovery probes)	RQ2 (embedding geometry)
MOMENT	61.8% avg balanced acc — middle	Loop DTW ~1.5 — dominant
TS2Vec e120	66.5% avg balanced acc — best	Loop DTW ~87 — worst

TS2Vec’s contrastive training pushes same-window representations together globally, which helps linear/RBF probes recover discrete factor labels — but the geometry is chaotic, loops don’t form, centroids drift. MOMENT’s pretrained features produce geometrically regular embeddings — tight, stable, fast mode transitions — without any task-specific training.

These are complementary strengths. TS2Vec for discrete label recovery; MOMENT for continuous online monitoring and anomaly detection.

1.13.2 13.2 MOMENT vs FFT: When Each Wins

Across all five RQs, two conditions predict which model wins on MSI:

Condition	MOMENT wins MSI	FFT wins MSI
Clean, periodic, sine signals	(RQ3 noise_000: 73.1 vs 37.2)	
High noise, periodic signals	(RQ3 noise_040: 59.9 vs 22.8)	
Chirp / damped / sawtooth signals	(RQ6: 51–57 vs 19–25)	
Mixed step+sine channels (1–3 step ch)	(RQ6: 48–73 vs 17–28)	
Pure step-change signals		(RQ3 sig_step: 9.6 vs 2.8)
Event/impulse signals		(RQ3 sig_event: 23.9 vs 18.7)
CNC machining, program steps		(RQ4: 45–82 vs 36–55)
CNC machining, operating states	~tied	~tied
Small window (ws=32)		FFT closer
Optimal window (ws=64)		FFT 64.3 vs MOMENT 48.6

MOMENT wins whenever the signal is periodic and smooth. FFT wins (or ties) whenever the signal has strong spectral structure that a simple amplitude spectrum captures completely, or when the signal is non-periodic. On geometry metrics (loop DTW, centroid stability), MOMENT wins **always and by a large margin** regardless of condition.

1.13.3 13.3 The Loop Consistency Invariant

Across all 5 RQs, MOMENT loop DTW is bounded between 1.25 and 11.1 regardless of noise level, missing data type, number of modes, signal type, window size, or scenario complexity. FFT loop DTW ranges from 66 (best case P1) to 2214 (cnc_two_parts op-state). This suggests MOMENT’s pretrained representation has a structural property that enforces temporal consistency in a way FFT does not — likely the transformer’s attention mechanism averaging over the full window context rather than computing a fixed spectral decomposition.

1.13.4 13.4 Production Risk Summary

Risk	FFT	MOMENT
Sensor dropout (channel missing)	Critical — MSI 13.5, loop DTW 478	Not measured — likely critical
High noise ($\sigma = 0.40$)	Moderate — MSI drops 38%	Low — MSI drops 18%
Step/event signal type	Moderate — MSI 9.6	Severe — MSI 2.8
Many modes (K=8)	Moderate — MSI 30.9	Low — MSI 52.2
Small window (ws=32)	Moderate	Low (padding robust at ws=64+)
Large window (ws $\geq$ 256)	Severe — MSI collapses	Moderate — more stable

1.14 14. Lessons Learned

1.14.1 14.1 Method Design

#	Lesson
L1	Temporal-neighbour contrastive pairs teach temporal proximity, not mode identity.
L2	Reconstruction loss adds complexity but doesn’t fix misaligned objectives.
L3	More training helps probes but can harm global structure.
L4	Pretrained foundation models give strong geometry without task-specific training.
L5	Load (continuous amplitude) is not learnable from the current setup — $R^2 \leq 0$ for all methods.
L6	MOMENT’s pretraining assumption fails on <i>purely</i> step/event signals — but mixed signals still favour
L7	Channel dropout is a critical production failure mode for all tested methods.
L8	MOMENT loop consistency is an invariant: ~ 0.002 centroid stability regardless of signal type, window s
L9	Exotic continuous signals (chirp, damped, sawtooth) favour MOMENT strongly — FFT MSI drops 49%

1.14.2 14.2 Evaluation Design

#	Lesson	Evidence
L8	Retrieval R@K is misleading when training biases nearest neighbours.	Phase 2: R@5 > 0.96 but p
L9	Single-seed results for TS2Vec should be treated cautiously.	MOMENT coupling std ± 0
L10	Geometry metrics reveal embedding quality that probe accuracy misses.	MOMENT dominates RQ2
L11	PCA compactness may penalise high-dimensional embeddings unfairly.	MOMENT PCA compact 1
L12	Window size must be smaller than the shortest mode segment duration.	RQ5: op-state MSI=0 at ws

1.14.3 14.3 Benchmark Design

#	Lesson
L13	FRS is a valid instrument for spectral/mode learning but doesn't capture step signals, discrete channels
L14	High baseline performance (FFT avg 60.3%) indicates strong spectral structure — learned methods m
L15	Missing data stress-testing is essential before production recommendation.
L16	Two-level label structure (coarse/fine) reveals that the same embedding cannot optimise both simultan

1.15 15. Open Questions

1. **Why does MOMENT fail so severely on step signals?** MSI 2.8 on sig_step while retaining

loop DTW 10.4 — the orbits are consistent but entirely overlapping. Is the T5 backbone averaging out the step transitions into a smooth mean?

1. **Can factor-aware pairs fix the load problem?** Windows with matching load values as positives

instead of temporal neighbours — the most direct fix for Load $R^2 \leq 0$.

1. **Does MOMENT's loop consistency invariant hold under channel dropout?** The miss_channel

scenario was not evaluated for MOMENT. If loop DTW is still low despite collapsed MSI, it has important implications for fault-tolerant monitoring architectures.

1. **What is the optimal ensemble?** RQ4 shows FFT and MOMENT have complementary strengths:

FFT for MSI / label classification, MOMENT for loop consistency / anomaly detection. A simple ensemble (MOMENT geometry for anomaly detection + FFT features for mode classification) might outperform either alone.

1. **Would DCC (contrastive training with factor-aware pairs) break the role reversal?**

The RQ3 DCC training jobs failed due to an import bug (now fixed). Running DCC would tell us whether a trained model can achieve both high probe accuracy (like TS2Vec) and good geometry (like MOMENT).

1. ****Why does MOMENT perform so much better in RQ6 ratio_4step (MSI 27.7) than RQ3 sig_step**

(MSI 2.8) if both are pure step-channel?** The mode-level step amplitude patterns in RQ6 likely differ across modes (each mode has a fixed level per channel), whereas RQ3 sig_step used level differences that overlapped. Confirms that the signal-type failure is amplitude overlap, not just aperiodicity.

1. **Does ws=64 generalise across scenarios?** RQ5 only tested one scenario (cnc_part_full).

The optimal window size likely varies with spindle speed — at high RPM, 64 samples may capture less than one vibration cycle.

1.16 16. What Remains To Be Done

1.16.1 Immediate

1. **RQ3 DCC training** — resubmit rq3_dcc.slurm (import bug fixed). Will add a trained

contrastive baseline to the RQ3 landscape sweep.

1. **Sync RQ3–5 results locally** — rsync results and plots from Alvis.
2. **RQ2 visualisation plots** — the viz step failed for RQ2 eval due to missing matplotlib; regenerate locally.

1.16.2 Track A — FRS Fixes (highest priority)

- **A1:** Factor-aware positive pairs for TS2Vec (fix load and coupling learning)
- **A2:** Load regression auxiliary head
- **A4:** Multi-seed TS2Vec to quantify variance
- **A3:** OOD splits (unseen factor combinations at test time)
- **A5:** Segment-level change-point detection evaluation

1.16.3 Track B — PLC Log Bridge

- **B1:** Synthetic PLC log corpus with FRS-like factors and mixed signal types
- **B2:** Tabular/numerical embedding extraction with step-aware encoding
- **B3:** Cross-modal comparison with FRS eval infrastructure

1.16.4 Track C — Geometry and Architecture

- Replace PCA compactness with silhouette score in the full embedding space
- Test MOMENT with fine-tuning on a small labelled set (few-shot regime)
- Investigate whether a lightweight adapter on MOMENT’s 768-dim output can recover

the Load factor that frozen MOMENT misses